

Meeting 4. The Cost of Being Wrong

Act 1 Scoring a Guess: Likelihood · Act 2 The Cross-Entropy Loss

About these notes. There is no textbook to buy in CS 301: these notes are the course’s text, written by the instructors and posted on Canvas as each meeting is reached. They follow the meetings themselves — including questions and examples that came up in class — and, together with the assignment notebook you build, they are what the exams are set from. Logistics, grading, and the schedule live in the syllabus.

Meeting 2 ended with a question that has been waiting ever since: you chose weights for AND by hand, and you wrote down how you did it — could something other than a person follow that procedure? Meeting 3 refused a smaller question. Shown a line that puts 707 of 6,865 days on the wrong side, we counted the wrong days and left it there.

Neither question can be answered until one thing exists: a way to say how good a candidate setting is, as a single number. A search that cannot compare two guesses cannot prefer one. This meeting builds that number. Nothing is searched here — every setting compared below was chosen by hand — but by the end the unit has a score, and a wrong answer has a price.

The names so far. Everything in this meeting is built from the unit as meetings 2 and 3 left it. The meeting opened by going over the review-quiz answers, and these were the names that needed repeating.

- the **score** $s = \mathbf{w} \cdot \mathbf{x} + b$: one number per input. The unit multiplies each input by its weight, adds the results, and adds b ;
- the **bias** $b = -\theta$, so that “fires when the sum reaches θ ” becomes “fires when $s \geq 0$ ”. Two names for one number: the threshold says how large the weighted sum must be, and the bias is that number with its sign reversed. The word *bias* fits because b is the unit’s own disposition: with a large b the unit fires at the slightest input, and with a very negative b it hardly fires at all;
- the **sigmoid** $\sigma(s) = 1/(1+e^{-s})$, which turns that score into a number strictly between 0 and 1;
- the **temperature** T , a width in the units of the score: $\sigma(s/T)$ is the output, and T says how far from the boundary an input must be before the unit is nearly sure. Changing T changes the probability and never the decision: a positive score divided by any positive T is still positive, so the day stays on the same side of the line;
- the **label** y , which is 0 or 1 and nothing else.

Vectors, once more. Several quiz answers said that meeting 2’s notation was the least clear part so far, so it was repeated in class, and it is repeated here. It is all the linear algebra Unit I uses.

The inputs $x_1, x_2, \dots$ are numbers, one per input. Written side by side they form a **row vector** $\mathbf{x}$. The weights $w_1, w_2, \dots$ form a row vector $\mathbf{w}$ in the same way. A vector can just as well be written as a column, and the **transpose**, written $\mathbf{x}^\top$, is the operation that turns a row into a column or a column into a row. The same operation applies to a matrix: each row of A becomes a column of $A^\top$, and each column becomes a row. One rule about it was stated in meeting 2 and is worth keeping: the transpose of a product is the product of the transposes in the reverse order, $(AB)^\top = B^\top A^\top$.

The score is the simplest case of matrix multiplication: a row times a column. Multiply the entries that stand in the same position and add the products,

$$\mathbf{x} \mathbf{w}^\top = x_1 w_1 + x_2 w_2 + \dots = \mathbf{w} \cdot \mathbf{x}.$$

Meeting 2’s notes wrote the two vectors as columns, so the same sum appeared there as $\mathbf{w}^\top \mathbf{x}$; in class it was written with rows, as above. Either way it is one number. The general rule, for matrices with many rows and columns, is the same rule applied row by row and column by column; it arrives in Unit II, by which point Math 337 will have covered it for those taking the two courses together.

Act 1. Scoring a Guess: Likelihood

1.1 Counting is a bad score

One word first, because it is about to be used in two ways. The unit computes a score for each input: $s = \mathbf{w} \cdot \mathbf{x} + b$. What this meeting builds is a score *for the unit*, computed by us, from the data, to say how well the unit does on it. The first is one number per day; the second is one number per setting of $\mathbf{w}$ and b .

The obvious score is the one meeting 3 used: count the days the line puts on the wrong side, and prefer the line with fewer. For the hand line high + low = 120 that count is 707 of 6,865, and it does not depend on the temperature, because dividing every score by T moves no day across the line. The count has two defects, and both matter.

It ignores confidence. Take two lines, both wrong about November 6, 2024. One gave that day a probability of 0.51 of being in the warm half; the other gave it 0.93. By the count they are equally wrong. They are not equally wrong: the second was confident and mistaken, which is worse than being mistaken while unsure. Read the 0.93 as the claim it is. The unit is not only saying “warm half” about that November day; it is saying that of a hundred days that look like this one, seven will be in the cold half. The day was in the cold half, so the unit put it among the seven. A unit that said 0.51 made the same wrong call while admitting it was nearly a coin toss, and that admission is worth something.

It barely moves. Shift a line a little and, almost always, no day crosses to the other side — the days sit at scattered points, and a small move of the boundary steps over none of them.

The count is therefore the same number for a whole neighbourhood of lines, and a score that cannot tell nearby candidates apart cannot guide a choice between them.

What is wanted instead is a score that reads the unit's probabilities rather than only its verdicts, and that changes whenever the setting changes.

1.2 What it means to explain the data

Meeting 3 read the unit's output as its confidence that the label is 1. There is a second reading of the same number, and it makes this meeting easier: **the output is the probability that the neuron fires**.

Take one row, with input $\mathbf{x}_i$, label y_i , and score s_i .

- If $y_i = 1$, the neuron *should* fire. The probability that it was right about this row is the probability that it fires: $\sigma(s_i)$.
- If $y_i = 0$, the neuron should stay silent. The probability that it was right is the probability that it does *not* fire: $1 - \sigma(s_i)$.

Write p_i for that number, whichever case the row falls into: p_i is **the probability that the neuron was right about row i** . Three days of the weather data, under the hand-drawn line $\text{high} + \text{low} = 120$ with $T = 10$:

day	high	low	label	p_i
July 10, 2024	93	79	warm half	0.995
November 6, 2024	83	63	cold half	0.07
May 9, 2020	50	34	warm half	0.027

A setting explains the data well when p_i is high on row after row. This setting explains the July day very well and the other two badly — which is the arithmetic form of what meeting 3 said about them.

1.3 One expression, both cases

The definition of p_i just given is a definition *by cases*: $\sigma(s_i)$ in one circumstance, $1 - \sigma(s_i)$ in the other. That is perfectly clear to a person reading one row at a time, and it is awkward everywhere else. A formula with the word “if” in it cannot be handed to a computer that is processing thousands of rows at once, and it cannot be manipulated algebraically as a single object.

So: is there one expression that gives $\sigma(s)$ when $y = 1$ and $1 - \sigma(s)$ when $y = 0$, with no cases at all?

There is, and it uses a fact that is easy to walk past: **y is 0 or 1 and nothing else**. A quantity that can only be 0 or 1 can be used as a switch. For any two numbers A and B , consider

$$yA + (1 - y)B.$$

When $y = 1$ this is $1 \cdot A + 0 \cdot B = A$. When $y = 0$ it is $0 \cdot A + 1 \cdot B = B$. One of the two terms is multiplied by 1 and survives; the other is multiplied by 0 and vanishes. The label chooses.

Now take $A = \sigma(s)$ and $B = 1 - \sigma(s)$:

$$p_i = y_i \sigma(s_i) + (1 - y_i)(1 - \sigma(s_i)).$$

This is the same rule as before — not an approximation of it, not a different rule that usually agrees — written without cases. It is worth being suspicious of, because it looks like an average of the two possibilities, and it is not: since y never takes a value between 0 and 1, nothing is ever blended. Exactly one term is ever alive.

Example 1.1 The label as a switch

Check both branches on numbers.

A row whose label is 1, with $\sigma(s) = 0.9$. The expression is $1 \cdot 0.9 + (1 - 1) \cdot (1 - 0.9) = 0.9 + 0 \cdot 0.1 = 0.9$. The neuron should fire and fires with probability 0.9, so it was right with probability 0.9. Correct.

A row whose label is 0, with $\sigma(s) = 0.3$. The expression is $0 \cdot 0.3 + (1 - 0) \cdot (1 - 0.3) = 0 + 0.7 = 0.7$. The neuron should stay silent, and it stays silent with probability 0.7, so it was right with probability 0.7. Correct — and notice that the unit's own output, 0.3, is not the answer. On a row labelled 0 the unit is doing well when its output is *small*.

One warning, for later. The term that gets multiplied by zero is still written down, and it can be a perfectly awful number: on the second row above, the dead term contains $\sigma(s) = 0.3$, and if $\sigma(s)$ had been 0 it would contain $\log 0$. In arithmetic, zero times anything is zero and the dead term is harmless. In a computer, $0 \times (-\infty)$ is not zero; it is **nan**, “not a number,” and it poisons everything it touches. §2.8 returns to this.

1.4 Likelihood

One row at a time is not yet a score for the whole dataset. Combine them.

If the rows are taken to be independent — each day's outcome does not depend on the others — then the probability that the neuron was right about *every* row is the product of the per-row probabilities. That number has a name:

$$\text{the likelihood of a setting} = \prod_{i=1}^n p_i.$$

Read it either way round. It is the probability that the neuron got the whole dataset right; equivalently, it is the probability of everything that actually happened, computed as though the unit's probabilities were the true ones. Higher is better, and it is one number for a whole setting, which is what §1.1 asked for.

Two consequences follow from its being a *product*, and they are the reason it behaves as a score should.

One confident mistake hurts. A row on which the neuron was right with probability 0.07 multiplies the whole likelihood by 0.07. No amount of being right elsewhere undoes that cheaply.

One certain mistake is fatal. A row with $p_i = 0$ sends the product to exactly 0. The unit of meeting 1, which answers only 0 or 1, is wrong on 707 days of the weather data and assigns each of them a probability of 0. Its likelihood on that data is exactly zero, and would remain zero if it were right about every other day in the record. This is what meeting 3 meant

by saying that the sigmoid unit, wrong about the same November day, at least “gave 7% to what happened.”

Example 1.2 Likelihood by hand

The unit with $w = (1, 1)$, $\theta = 1.5$ and $T = 1$, asked for AND. This is the first task of the meeting’s sheet.

The score is $s = x_1 + x_2 - 1.5$, so the four rows give -1.5 , -0.5 , -0.5 and 0.5 . Applying the sigmoid: $\sigma(-1.5) = 0.18$, $\sigma(-0.5) = 0.38$, $\sigma(0.5) = 0.62$.

Now the switch. The first three rows are labelled 0, so their p_i is $1 - \sigma$; the last is labelled 1, so its p_i is σ .

x_1	x_2	AND	s	$\sigma(s)$	p_i
0	0	0	-1.5	0.18	0.82
0	1	0	-0.5	0.38	0.62
1	0	0	-0.5	0.38	0.62
1	1	1	0.5	0.62	0.62

The likelihood is $0.82 \times 0.62 \times 0.62 \times 0.62 = 0.20$.

Now hold the same line more firmly: double the weights and the threshold, to $w = (2, 2)$ and $\theta = 3$. The line is unchanged — $2x_1 + 2x_2 = 3$ is the same set of points as $x_1 + x_2 = 1.5$ — but the scores are twice as large: $-3, -1, -1, 1$. The probabilities become $0.95, 0.73, 0.73, 0.73$, and the likelihood rises to 0.37 .

1.5 On a truth table, certainty is free

Continue the doubling. Multiplying w and θ by k leaves the boundary exactly where it was and multiplies every score by k :

k	p_i on the four rows				likelihood
1	0.82	0.62	0.62	0.62	0.20
2	0.95	0.73	0.73	0.73	0.37
4	0.9975	0.88	0.88	0.88	0.68
8	1.0000	0.98	0.98	0.98	0.95

Every row improves, every time. The reason is worth stating plainly: on this table every row is already on the correct side of the line, so every row is a row where more confidence means more credit. Nothing pulls the other way, and the likelihood climbs toward 1 as k grows.

Hold on to the phrasing — *every row is on the correct side* — because it is exactly what fails on real data, and §2.5 is about what happens then.

1.6 A product of 6,865 numbers

The likelihood has one practical defect, and it is severe. On the weather data it is a product of 6,865 numbers, each smaller than 1. For the hand-drawn line at $T = 10$ its value is about 10^{-730} .

A double-precision floating-point number — the kind Python uses — cannot represent anything smaller than about 10^{-308} . The likelihood of this perfectly ordinary setting is a real

number that the machine silently rounds to zero. Two different settings, one much better than the other, would both be reported as zero, and the score would be useless exactly where it was needed.

The fix is one line of high-school algebra, and it is the subject of the second act.

Act 2. The Cross-Entropy Loss

2.1 From a product to a sum

The logarithm turns products into sums: $\log(ab) = \log a + \log b$, and so

$$\log \prod_{i=1}^n p_i = \sum_{i=1}^n \log p_i.$$

This is the quantity called the **log-likelihood**. Two things make it a safe substitute for what it replaces.

It ranks settings identically. The logarithm is an increasing function, so if one setting has the larger likelihood it has the larger log-likelihood. Nothing about which setting is preferred can change.

And it is a number the computer can hold. Each $\log p_i$ is a modest negative number, and their sum for the weather at $T = 10$ is about $-1,680$: unremarkable to store, where 10^{-730} was impossible.

Throughout this course the logarithm is the **natural** logarithm, written $\ln$ or $\log$ interchangeably here. The choice matters only for the units, which §2.7 will name.

2.2 From goodness to loss

Two conventions convert the log-likelihood into the quantity actually used.

Flip the sign. The log-likelihood is negative, and larger is better; it is easier to work with a quantity that is positive, and where the goal is to make it *small*. Negating does that. The name for such a quantity is a **loss**.

Divide by the number of rows. A truth table has 4 rows and the weather has 6,865, so their totals are not comparable; per-row averages are. This also means that a loss can be quoted for a dataset without saying how big the dataset is.

Applying both to the log-likelihood, and substituting the switch form of p_i from §1.3, gives the **cross-entropy loss**:

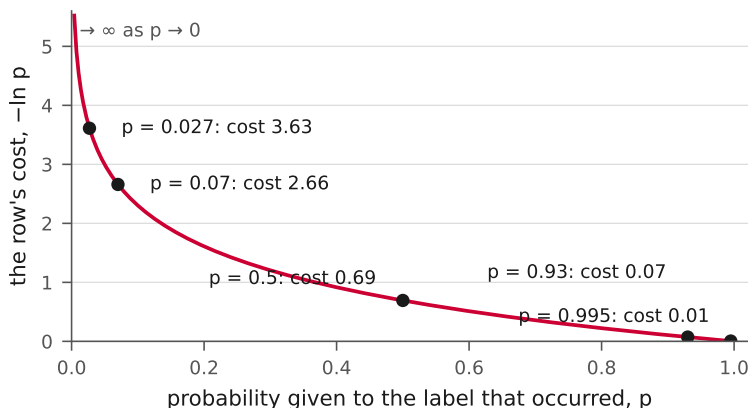
$$L = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log \sigma(s_i) + (1 - y_i) \log (1 - \sigma(s_i)) \right].$$

The bracket deserves a second look, because it is the switch of §1.3 again, now inside the logarithm rather than outside it. There, one of two *probabilities* was selected; here, one of two *logarithms* is. On a row with $y_i = 1$ the bracket is $\log \sigma(s_i)$, and on a row with $y_i = 0$ it is $\log(1 - \sigma(s_i))$ — in both cases $\log p_i$, which is what the sum needs.

This step is safe for the same reason as before, and only for that reason: y is 0 or 1. It is not generally true that $\log(yA + (1 - y)B)$ equals $y \log A + (1 - y) \log B$; with, say, $y = \frac{1}{2}$ the left side is the logarithm of an average and the right side is an average of logarithms, and those differ. What rescues the identity is that y never takes such a value. At $y = 1$ both sides are $\log A$; at $y = 0$ both are $\log B$; and there is no third case to check.

2.3 One row's contribution

Each row contributes $-\log p_i$ to the loss. It is worth seeing that function on its own.



The curve is the price of being wrong, read against how much probability the unit gave to the truth.

- $p = 0.995$ costs 0.01. The unit was sure and right: almost free.
- $p = 0.93$ costs 0.07; $p = 0.5$ costs 0.69.
- $p = 0.07$ costs 2.66 — November 6, 2024 under the hand line.
- $p = 0.027$ costs 3.63 — May 9, 2020.
- And as p approaches 0, the cost grows without bound.

This is what it costs that the sigmoid never quite reaches 0 or 1. A wrong answer given with confidence costs a finite amount, and the more confident it was, the more it costs. A wrong answer given with *certainty* costs infinitely: the 0-or-1 unit, on that same November day, pays a contribution no number of correct days can offset. This is the sense in which this meeting's title should be read: the loss is the price of being wrong, and how confident the unit was sets the price.

Example 2.1 Cross-entropy by hand

The two settings of Example 1.2, now scored by the loss. This is the second task of the meeting's sheet.

For $k = 1$ the probabilities were 0.82, 0.62, 0.62, 0.62. Their negative logarithms are 0.20, 0.48, 0.48, 0.48, and the loss is the average:

$$L = \frac{0.20 + 0.48 + 0.48 + 0.48}{4} = \frac{1.64}{4} = 0.41.$$

For $k = 2$ the probabilities were 0.95, 0.73, 0.73, 0.73, so the contributions are 0.05, 0.32, 0.32, 0.32 and the loss is $1.01/4 = 0.25$. Doubling the weights halved the price, roughly, and it did so without moving the line by a hair.

Two mistakes are worth naming because they are the common ones. Dropping the minus sign gives a negative loss, which is impossible: every contribution is the negative logarithm of a number below 1, hence positive. And skipping the division by 4 gives a number that cannot be compared with anything computed on a dataset of a different size.

2.4 Why the loss can reach zero on a truth table

Take the doubling further. The likelihoods of \$1.5 correspond to these losses:

k	1	2	4	8
likelihood	0.20	0.37	0.68	0.95
loss	0.41	0.25	0.096	0.014

The loss falls toward zero, and the word “toward” is exact: since σ never attains 1, no p_i is ever exactly 1, no contribution is ever exactly 0, and the loss is never exactly zero. But there is no lower bound short of zero either — name a small number and a large enough k gets under it.

This is a fact about truth tables, not about units. A truth table gives each input exactly one output, the same output every time, and the four rows of AND happen to be arranged so that a line can put all of them on their correct sides. Once that is true, confidence is free, and the price of being wrong can be driven to nothing because the unit is never wrong.

2.5 And never on the weather

On the weather data the same manoeuvre stops working, and it is worth seeing exactly where.

There are 6,865 days and 1,878 distinct (high, low) pairs, so many days share an input with some other day. Of those pairs, **337 occur with both labels** — the same high and the same low, once in the warm half of the year and once in the cold. Those pairs cover 1,881 days, more than a quarter of the data.

Take one of them. A high of 82°F with a low of 63°F occurs on 14 days: **12 in the warm half and 2 in the cold half**. The unit sees the same two numbers on all 14 days, so it returns the same probability on all 14 days — it has nothing to distinguish them with. Call that probability q . Then 12 days pay $-\ln q$ each and 2 days pay $-\ln(1 - q)$ each, and the average contribution over the 14 is

$$\frac{12}{14}(-\ln q) + \frac{2}{14}(-\ln(1 - q)).$$

Choosing q to make this as small as possible gives $q = 12/14 = 0.86$ — believe what the data does — and a cost of 0.41 per day. Not zero. Choosing any other q makes it worse, and choosing q close to 0 or 1 makes it far worse: the two cold days alone would pay $-\ln(1 - q)$ each, which grows without bound as q approaches 1.

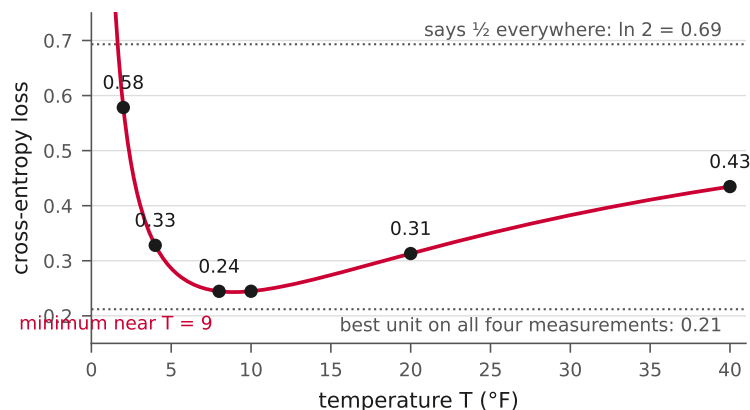
So there is a floor, and no setting can get under it. Here is the scale of the whole thing, for the weather:

a unit that answers $\frac{1}{2}$ on every day	0.693
a clearly wrong line, high + low = 160	1.11
the hand-drawn line at its best T	0.243
the best line on (high, low)	0.217
the best unit on all four measurements	0.212

The floor is a little above 0.21, and it is a property of the data. April and October have nearly the same weather and opposite labels; no rule computed from four measurements can separate days that the measurements do not distinguish. Meeting 3 put it as *the question is hard*; this is the same statement with a number attached.

2.6 Temperature, seen by the loss

The loss also settles a question meeting 3 left open. There, the value $T = 10$ was set by hand, with no reason given. With a loss in hand, the data can be asked which T it prefers. Keeping the hand-drawn line fixed and changing only T :



There is a minimum, near $T = 9$. Sharpening the unit below that value makes it worse, and blurring it above that value makes it worse.

The reason is the competition §1.5 said was missing on a truth table. Lowering T makes every output more extreme. On the roughly 6,158 days that the line places correctly, that lowers its contribution. On the 707 days it places wrongly, that raises the contribution — and raises it steeply, since those rows sit on the expensive part of the curve in §2.3. The total falls while the first effect dominates and rises once the second does, and between them sits a best sharpness. On a truth table there is no second effect, which is why the same curve there just keeps falling.

One consequence, worth noticing now: multiplying the weights and the bias by k has the same effect as dividing T by k , since both multiply every score by the same factor. Sharpness is not a separate thing from the size of the weights. They are one quantity wearing two names.

2.7 Why it is called cross-entropy

The loss was built here out of a probability, a logarithm and an average, and nothing about that construction explains its name. The name is borrowed, and the thing it is borrowed from is worth a page.

Start with a single number: if something you believed had probability p happens, then $-\log p$ measures how **surprised** you should be. If you were certain ($p = 1$) the surprise is 0. The less probability you gave it, the more surprised you are, without limit. That is precisely the contribution of §2.3 — so the cross-entropy loss is the *average surprise* of the unit, over the rows, at what actually happened. A unit that explains the data is a unit that is rarely surprised by it.

Now suppose outcomes occur according to some distribution p — outcome i happening a fraction p_i of the time. The average surprise of someone who believes p and sees the world do p is

$$H(p) = - \sum_i p_i \log p_i,$$

and this quantity is the **entropy** of p . It was defined by Claude Shannon in 1948, in the paper that created information theory, and it measures how uncertain the situation is in the first place: a coin that always lands heads has entropy 0 and there is nothing to learn; a fair coin has the largest entropy two outcomes can have.

Entropy has a unit, and the logarithm chooses it. With logarithms base 2 the unit is the **bit**, and the entropy of a fair coin is exactly 1 bit — one yes-or-no question settles it. With natural logarithms, as here, the unit is called the *nat*, and the same fair coin has entropy $\ln 2 = 0.693$ nats. That number should look familiar: it is the loss of a unit that answers $\frac{1}{2}$ on every day, from the table in §2.5. Such a unit has learned nothing from the measurements, so each day still costs a full yes-or-no question. It is not a coincidence; it is the same quantity.

Finally the “cross”. Suppose the world does p but you believe some other distribution q . Your average surprise is then

$$H(p, q) = - \sum_i p_i \log q_i,$$

the **cross-entropy** of q relative to p : the outcomes are weighted by how often they really happen, but the surprise of each is computed from what you believed. It is smallest when $q = p$, and every departure of your beliefs from the truth costs you something.

Our loss is exactly this, one row at a time. On a row, what the world does is not in doubt — the label happened — so the true distribution puts all of its weight on the observed label and none on the other. What the unit believes is $\sigma(s_i)$ for label 1 and $1 - \sigma(s_i)$ for label 0. Substituting those into the formula above leaves a single surviving term, $-\log p_i$, which is the row’s contribution; averaging over rows gives L . The loss is the average cross-entropy between what happened and what the unit believed, and the name is a description rather than a decoration.

The last line of §2.5 can now be said in bits. A unit that says $\frac{1}{2}$ everywhere pays 1 bit per day. The best unit on the four measurements pays 0.212 nats, which is 0.31 bits. The difference, about two-thirds of a bit per day, is what a day’s weather is worth when the question is which half of the year it fell in.

C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal 27 (1948), 379–423 and 623–656. Shannon appears in this course twice: his 1937 master’s thesis, *A Symbolic Analysis of Relay and Switching Circuits*, showed that Boolean algebra describes switching circuits, which is the reason the machine of meeting 1 is built out of logic gates at all. The principle of preferring the setting that makes the observed data most probable is older still, and is due to R. A. Fisher in the 1920s.

Beyond the lecture: cross-entropy in general. Nothing below is needed for this unit, and none of it is examined here. It is included because the quantity is much more general than the use made of it above, and the general form is the one you will meet elsewhere.

Cross-entropy is defined for any two distributions over the same outcomes, and there may be any number of outcomes rather than two. If p is what happens and q is what is believed, $H(p, q) = -\sum_i p_i \log q_i$ as above. Three properties are worth knowing.

It is never smaller than the entropy. $H(p, q) \geq H(p)$, with equality exactly when $q = p$. Believing the truth is optimal, and believing anything else costs more.

The excess has its own name. The gap $H(p, q) - H(p)$ is the **Kullback–Leibler divergence** $D(p \parallel q)$, the part of the cost that is your fault rather than the world's. So $H(p, q) = H(p) + D(p \parallel q)$: total surprise equals unavoidable uncertainty plus penalty for being wrong. In our per-row use the true distribution is concentrated on one outcome, so $H(p) = 0$ and the cross-entropy is *all* penalty — every nat of the loss is the unit's own.

It measures message length. Shannon's setting was communication. If outcomes occur with distribution p and you encode them with a code designed for q , the average length of your messages, in bits, is $H(p, q)$; the shortest possible average is $H(p)$, achieved by coding for the truth. This is why the loss has the units it does, and it is the sense in which a model that predicts well is a model that compresses well.

The floor of §2.5 is this idea applied backwards. Group the days that share an input: at (82°F, 63°F) the label is warm on 12 of 14 days, so the distribution of the label *given* that input is (12/14, 2/14), whose entropy is 0.41 nats — exactly the cost computed there. The unit cannot beat the entropy of the label given what it is allowed to see. Averaged across a dataset this quantity is called the conditional entropy of the label given the features, and it is the true floor for any model whatever, linear or not. Estimating it from data is harder than it looks — most (high, low) pairs in this record occur only a handful of times, and a pair seen once always looks perfectly predictable — so trust it where the counts are large, as at that one pair, and treat the fitted numbers in §2.5 as the practical statement.

2.8 Computing it without breaking it

Written directly, the loss is two lines of NumPy:

```
p = sigmoid(s)
loss = -np.mean(y * np.log(p) + (1 - y) * np.log(1 - p))
```

On the AND table this reproduces the 0.41 and 0.25 of Example 2.1, and on the weather with the hand line at $T = 10$ it gives 0.245.

Run the same two lines at $T = 1$, however, and the result is `inf` — although the true loss at $T = 1$ is 1.12, a perfectly ordinary number. Two facts from earlier meet here. At $T = 1$ the scores are ten times larger, and 804 of the days have a score above about 37. The sigmoid of 37 is not 1, but it is closer to 1 than a double-precision number can express, so it *is stored as* exactly 1.0 — the finite-precision remark of meeting 3, arriving with consequences. Then $\log(1 - 1.0) = \log 0 = -\infty$, one contribution is infinite, and the average of anything with an infinity in it is an infinity.

Worse, the warning in Example 1.1 now matters: on such a row the infinite logarithm may be the term multiplied by $y = 0$, and $0 \times (-\infty)$ evaluates to `nan`, not to 0. The formula is

correct; evaluating it in this order is not.

The cure is to never form p at all. Writing z for the score divided by the temperature, a little algebra gives

$$-\log \sigma(z) = \log(1 + e^{-z}), \quad -\log(1 - \sigma(z)) = \log(1 + e^z),$$

so each contribution is $\log(1 + e^x)$ for a suitable x . That function is computed safely as

$$\log(1 + e^x) = \max(x, 0) + \log(1 + e^{-|x|}),$$

where the exponential is now of a negative number and cannot overflow. NumPy provides it as `np.logaddexp(0, x)`, and the whole loss becomes

```
loss = np.mean(np.where(y == 1, np.logaddexp(0, -z), np.logaddexp(0, z)))
```

which returns 1.12 at $T = 1$ and agrees with the direct version everywhere the direct version works.

2.9 Two things to take with you

A guess can now be scored. For any setting, on any dataset, there is one number saying how well it explains what happened — computable by hand on four rows, and by one line of code on 6,865 days. It reads the unit's probabilities rather than just its verdicts, and it moves whenever the setting moves, which is what §1.1 asked for and what counting could not deliver.

And the two datasets behave differently, permanently. On a truth table the loss can be driven as close to zero as you like, because the unit can be right about every row and confidence is then free. On the weather it cannot go below about 0.21, because days that look identical carry different labels and no setting can tell them apart. That number is not a criticism of the line; it is the entropy that remains in the label once the measurements are known.

What still cannot be done is to choose the next guess. Meeting 2 asked whether a procedure could find the weights; this meeting has built the thing such a procedure would need, and has not built the procedure.

After this meeting.

- Assignment Task 2, *From certainty to confidence*: the sigmoid unit as a class, with **predict**, **accuracy** and **loss**, on both threads — the truth tables, where the loss goes toward zero, and the weather, where it does not. Find the surprising days in code, and report the loss of a hand-chosen unit on each dataset.
- Work both tasks of the meeting's sheet if you did not finish them: the likelihood first, then the same two settings scored by the loss.
- The week's review quiz is on Canvas.